

K-Bound: When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

Test-time adaptation updates a model without new target labels, but an update can help or harm. We ask when the available evidence supports replacing a frozen predictor with a fixed adapted candidate. K-Bound answers this question for a binary population model under an exogenous bound β on target calibration residual. We construct target label laws that preserve all label-free observations and derive the exact feasible interval for paired population benefit. A strict decision is justified precisely when $|M| > \beta$, where M is the observable correctness margin on the disagreement region; inside the closed band $|M| \leq \beta$, no strict direction is uniformly justified. Furthermore, an evidence-fibre audit theorem characterizes the smallest residual bound that any uniformly valid audit based on the same observable law must cover at its declared confidence level; additional processing of unchanged evidence cannot reduce the unresolved residual range. Addressing this problem under a complementary inductive regime with historical shift benchmarks, Knowability-Guided Adaptation (KGA) learns to predict observed evaluation-cell benefit, choosing ADAPT, FREEZE, or ABSTAIN via calibrated residual intervals. On cross-fitted CIFAR-10-C, Tent and EATA achieve lower point regret than either fixed policy, while SAR favors always-adapt due to high baseline benefit under noise. On real-camera natural shift (CCT-20), KGA served the frozen model across all 45 cells, avoiding the measured degradation of always-adapt without issuing any ADAPT decisions. These results demonstrate conservative retention in this setting and candidate-specific commitment control on synthetic shifts, while showing that selective opportunistic routing on unseen natural shifts remains an open empirical challenge.

Keywords: test-time adaptation; abstention; distribution shift; label-free risk estimation; conformal calibration; selective decisions.

1 Introduction

Test-time adaptation changes a model using unlabeled deployment data. Entropy minimization, filtering, regularization, and related methods can recover accuracy under shift, but the same update can also make predictions worse [1, 4, 5, 6]. An adaptation method proposes a candidate; a deployment system must still decide whether to use it.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

A positive benefit favors the update; a negative benefit favors the frozen model. We distinguish three conclusions: ADAPT supports positive benefit, FREEZE supports negative benefit, and ABSTAIN records insufficient evidence for either strict direction. Both non-adapt actions retain f_0 , but only freeze asserts that the candidate is worse.

K-Bound studies this decision as a partial-identification problem. In a binary model, we hold the input law, predictors, and label-free evidence fixed and vary the target labels within a declared class. The resulting feasible benefit set gives an exact boundary for a uniformly justified strict decision. The bound on the hidden calibration residual is an assumption about that class, not a quantity that an unlabeled batch supplies for free. A second result makes this information limit explicit: an audit using the same evidence must still cover the unresolved residual range.

We present K-Bound and Knowability-Guided Adaptation (KGA) together because they address the identical practitioner problem—when to risk model adaptation without ground-truth target labels—under two complementary evidentiary regimes. K-Bound operates in the deductive regime

of a single unlabeled test stream, establishing the fundamental impossibility of auditing target risk without an exogenous calibration sensitivity parameter β . KGA operates in the inductive regime where historical benchmark shifts are available, fitting an empirical benefit predictor on past evaluation cells and calibrating its residual uncertainty to govern a three-way commitment gate. We do not claim KGA implements or approximates the population frontier; rather, K-Bound defines what is fundamentally impossible from unlabeled test data alone, and KGA demonstrates what becomes operational once historical shift benchmarks are admitted into the evidentiary basis. Section 6.1 formalizes this architectural relationship.

avorable evidence.

1.1 Three Scientific Contributions

1. **An exact benefit frontier on a declared class.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **An evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **An empirical paired-benefit gate.** We give a reproducible three-way interval procedure and evaluate its commitment behavior under controlled and natural shifts. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The controlled CIFAR-10-C results show useful routing for some candidates, not a universal improvement: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt due to high baseline benefit under noise corruptions. On real-camera natural shift (CCT-20), KGA served the frozen model across all 45 cells and avoided the measured loss of always adapting, but selected ADAPT zero times. This demonstrates retention in that study, not general harm avoidance or successful selection of beneficial updates on unseen environments, confirming that selective opportunistic routing under unseen natural shifts remains an open empirical challenge. The supplementary results retain adverse, incomplete, and withheld studies, so this distinction does not depend on discarding unfav-

KGA decision flow

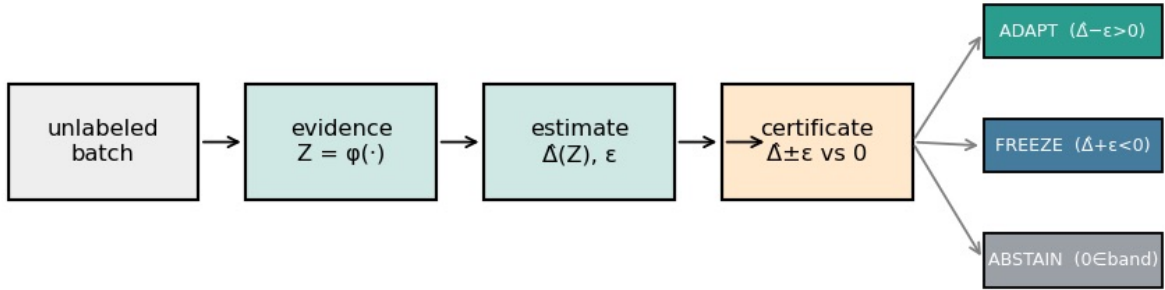


Figure 1: KGA’s empirical workflow. A shadow candidate proposes an update; a learned benefit prediction and residual interval determine whether to accept it. Historical labels support development and calibration, while the new decision uses no evaluation labels.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training improve candidate construction through entropy objectives, sample filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks [1, 4, 5, 6, 7, 2, 3]. Selective methods also choose where or when to update. TTSA selects modality-specific adapters [10]; GALA selects layers and filters samples using gradient alignment [11]; Informed Adaptation learns topological features associated with useful updates [9]. KGA evaluates a fixed proposed candidate against its frozen baseline. Learned routing and rollback are established ideas; their existence is not our novelty claim.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information [15, 17, 16, 18, 35, 34]. The closest empirical comparison is Kim et al.’s NeurIPS 2024 work, *Test-Time Adaptation Induces Stronger Accuracy and Agreement-on-the-Line* [51]. It uses agreement-based ALine estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA instead fits the paired observed benefit of

one frozen/candidate pair and uses an interval to make a three-way decision. This is a difference in the fitted target and decision procedure, not a claim that accuracy estimates cannot be differenced to compare models. In Section 8.3 (Table 9), we compare current KGA with fixed policies, scalar evidence rules and a point gate sharing its exact predictions. Historical TTA/ALine, AETTA and POEM port labels do not establish faithful implementations of the published estimators or controllers. Authenticated native comparisons under a shared protocol remain distinct from this saved-feature diagnostic.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting [14]. Schirmer et al. monitor the running risk of an adapting model relative to a source-risk reference, under a proxy-informativeness condition [8]; the supervised precursor uses a labeled test stream [13]. Drift-to-Action Controllers use delayed labels and online certificates to gate adaptation, abstention, rollback, and retraining [50]. These methods differ in the risk target, label access, and sequential guarantee. KGA concerns one paired benefit and does not establish an anytime-valid risk guarantee.

2.4 Partial Identification and Robust Decisions

An identified set describes the values compatible with the observations and assumptions. Decision theory under partial identification studies how to act when that set leaves payoffs ambiguous; Christensen et al. combine worst-case payoff uncertainty with statistical learning of identified parameters [54]. K-Bound addresses the narrower question of a uniformly correct strict benefit sign. It does not solve their general optimal-decision problem or claim that interval thresholding itself is new.

Indistinguishable adaptation problems are already central to domain-adaptation impossibility [20, 49]. K-Bound’s specific contribution is a disagreement-conditional label-kernel construction, its clipped benefit set, and the associated residual-audit floor. The proof does not introduce label-free impossibility. Risk-estimation assumptions based on conditional independence [19], label shift [44, 45], or disagreement discrepancy [21] restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure [12]. Such restrictions can change the identified set; they are not contradicted by a converse theorem for a richer class.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions [22, 46, 47]. KGA abstains on an update. Its coverage-to-action proposition applies standard interval logic [36, 37, 40, 41]. The main statistical work is to justify coverage, especially after selection or under shift.

Jin and Ren’s JOMI procedure targets coverage conditional on a unit being selected, under exchangeability and a compatible selection rule [52]. Gibbs et al. provide conditional guarantees over specified classes of shifts using a different calibration construction [53]. KGA’s ordinary residual interval implements neither procedure. Its marginal false-adapt bound therefore must be distinguished from error conditional on ADAPT and from coverage within each target environment. Learn Then Test separately addresses configuration selection and multiplicity [41].

Table 1: Closest roles, targets, and remaining distinctions. Literature comparisons are conceptual; the paper does not report a synchronized official head-to-head experiment.

Approach	Main target	Relation to this paper
TTA/ALine; AETTA	adapted-model performance and selection	closest empirical estimators; their scores can inform paired comparisons
POEM; risk monitors	protected adaptation or running risk	different monitored target and sequential assumptions
Partial-identification decisions	uncertain payoffs across compatible laws	context for the population benefit set and robust sign decision
Selection/shift-conditional conformance	coverage after selection or within a shift class	stronger, specialized targets not supplied by KGA’s marginal interval
K-Bound / KGA	feasible population benefit / predicted cell benefit	shared update question, distinct uncertainty objects and guarantees

3 Setup, Notation, and Claim Levels

3.1 Model Pair, Evidence, and Evaluation Unit

Let P_S and P_T be the source and target laws and $R_T(f) = \mathbb{E}_{(X,Y) \sim P_T}[\ell(f(X), Y)]$. Population risks integrate a fresh draw from P_T , holding the realized model pair fixed. An adapter $\mathcal{A}$ proposes f_a from f_0 and unlabeled inputs $X_{1:m}$. The decision can read

$$Z = \phi(X_{1:m}, f_0, f_a),$$

including predictions, logits, and update statistics, but not the new evaluation labels.

For evaluation unit j , with records $\mathcal{E}_j$ and a fixed higher-is-better score S , define

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

The two models are scored on the same records. This measured quantity is not population risk. For accuracy it is a sample average of paired correctness differences; for macro-F1 it is a difference of dataset-level scores, not an average of per-example losses. Metric scales are reported separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics). ADAPT commits f_a when the interval supports positive benefit. FREEZE rejects the candidate when it supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection. Abstention is an unresolved assessment. An operator can attach monitoring, label acquisition, or human review to the abstention record; the experiments here evaluate only the frozen fallback and uncertainty log, not those additional interventions. Retaining f_0 after missing evidence must be logged as ABSTAIN, not as a certified FREEZE.

The observed-cell oracle chooses the larger score. Regret is the policy’s score shortfall from that oracle, averaged over the named units. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA uses the candidate score only after ADAPT. Lower regret is better, and a zero-benefit tie carries zero regret for either model even though it supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

These are three different measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\text{ADAPT}, \text{FREEZE}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon), \quad (3)$$

$$\text{FA}_{\text{u}} = \mathbb{P}(g = \text{ADAPT}, \Delta^{\text{cell}} \leq 0). \quad (4)$$

Conditional false adaptation is $\text{FA}_{\text{c}} = \mathbb{P}(\Delta^{\text{cell}} \leq 0 \mid g = \text{ADAPT})$, defined only when $\mathbb{P}(g = \text{ADAPT}) > 0$. The operational proposition below bounds the marginal error under interval coverage; it does not bound FA_{c} . A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. Analogously, a false FREEZE is a FREEZE decision with $\Delta^{\text{cell}} \geq 0$; its conditional frequency divides by FREEZE decisions and is undefined when that action is absent.

Table 2: Notation and probability targets. Population and measured-cell quantities are kept separate.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D, \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
γ, β	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\hat{\Delta}, \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

Table 3: Three claim levels and their validity obligations. This table collects the distinctions used throughout the paper.

Claim level	Required premise	Conclusion and boundary
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class, not a learned residual bound
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target, not automatic conditional or anytime control
Empirical evaluation	fixed recorded candidates, splits, metric and action rule	measured cell regret, interval inclusion and action exposure; population transfer needs a separate argument

The population theorem uses binary labels and 0/1 loss. The empirical gate can instead target any predeclared scalar score difference, including multiclass metrics. This interface does not extend the binary theorem. In general multiclass classification, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, with $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$; only the binary setting forces $p_0 = 1 - p_a$.

4 Population Partial-Identification Frontier

The result describes which benefit signs are compatible with a fixed label-free evidence law and a declared bound on the hidden residual.

4.1 Disagreement-Region Model

Assumption 1 (Deployment setup). Let $Y \in \{0, 1\}$, use 0/1 loss, fix measurable binary predictors f_0, f_a , and assume $\mu_T(D) > 0$ for $D = \{x : f_0(x) \neq f_a(x)\}$. Let $s : \mathcal{X} \rightarrow [0, 1]$ be a fixed measurable candidate-correctness score, fitted using source information, and let the target law admit the measurable pointwise correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid$

$X = x)$. On D , set

$$M = \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \quad \gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]. \quad (5)$$

The declared population class is $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$ for an externally supplied $\beta \geq 0$. Because $s \in [0, 1]$, the feasible margin satisfies $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities and frontier below do not require a source-calibration property of s .

Remark 1 (Residual identity). The definition gives

$$M + \gamma = (\mathbb{E}[s(X) \mid D] - \frac{1}{2}) + \mathbb{E}[\eta_a(X) - s(X) \mid D] = \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}.$$

This is an identity, not empirical evidence that calibration transfers. We call γ a target disagreement-conditional calibration residual, not automatically source-to-target drift. Ordinary source calibration need not hold after conditioning on D , so γ can be nonzero even when $P_T = P_S$. A literal label-kernel shift interpretation would require an additional reference, for example the pointwise source correctness $s(x) = \mathbb{P}_S(f_a(X) = Y \mid X = x)$. The no-shift counterexample is given in the supplementary theorem-dependency map. Calibration notions also depend on the conditioning information: top-label calibration, for example, additionally conditions on

the predicted class [43]. Neither score nor top-label calibration automatically supplies calibration conditional on D .

Definition 2 (Risk alignment). Evidence Z is risk-aligned on a class $\mathcal{P}$ if no two laws in $\mathcal{P}$ with opposite nonzero benefit signs induce the same law of Z . This is weaker than uniform strict-sign identification: a fibre containing zero and positive benefit is risk-aligned by this definition, but does not support a uniformly positive claim. Population frontier claims take M to be fixed on each augmented evidence fibre, equivalently by using $\tilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is uniformly directionally sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$; FREEZE is sound when $\Delta(P) < 0$ for every such law. A pointwise maximal three-way rule commits whenever one strict action is sound and abstains otherwise. The strict convention treats $\Delta = 0$ as insufficient for either directional claim, although the actions then have equal task risk.

4.2 Label-Kernel Freedom on the Disagreement Region

Lemma 1 (Label-kernel freedom on disagreement). *Fix $(\mu_T, P_S, f_0, f_a, s)$, hence D and M , and fix the conditional label law outside D . For every measurable $\eta : D \rightarrow [0, 1]$ there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D . Every label-free observable formed from the input batch, the fixed models, and their outputs has the same law for all such η , while*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

Proof. On D , binary predictors disagree, so their outputs are 0 and 1. Assign conditional mass $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$, retaining the fixed kernel outside D . This changes no input marginal or model output and therefore changes no label-free evidence law. The expression for γ follows from Eq. (5). $\square$

4.3 Benefit Reduction and Matched-Evidence Impossibility

Lemma 2 (Disagreement-region reduction). *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a=Y \mid D)$,*

$$\begin{aligned} \Delta &= 2\mu_T(D)(\bar{a} - \tfrac{1}{2}), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \tfrac{1}{2}) = \text{sign}(M + \gamma). \end{aligned}$$

The identity $\text{sign } \Delta = \text{sign}(M + \gamma)$ holds for every target law; it does not assume $|\gamma| \leq \beta$.

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ ,

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}, \end{aligned}$$

so $\text{sign } \Delta = \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma)$. $\square$

Theorem 1 (Interior matched-evidence impossibility). *Fix $\beta > 0$. For any feasible margin $M \in [-\frac{1}{2}, \frac{1}{2}]$ with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that $\text{Law}(\tilde{Z} \mid P_T^1) = \text{Law}(\tilde{Z} \mid P_T^2)$ for $\tilde{Z} = (Z, M)$, $M(P_T^1) = M(P_T^2) = M$, and $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\text{sign } \Delta = \text{sign } M$ is determined and no such pair exists: matched-evidence ambiguity requires a positive declared residual bound.)*

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$, holding the off- D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ , and: *class membership*, $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \leq \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence*, $\text{Law}(W \mid P_T^\eta)$ does not depend on η by the lemma and $\tilde{Z} = (Z, M)$ is a function of W , so $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$; *opposite signs*, Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\theta\delta \neq 0$ with $\text{sign } \Delta_\theta = \theta$. $\square$

The two evidence laws here coincide *exactly*, not merely approximately: the construction changes only the label kernel on D , and Z is a function of unlabeled inputs and model outputs. We therefore do not present this as a Le Cam two-point argument. A Le Cam argument trades off a positive total-variation distance against the achievable error; here the distance is zero, which is the degenerate and strongest case, and the corresponding statement is exact non-identifiability rather than a rate.

Corollary 1 (Matched-evidence abstention lower bound). *Fix $\alpha \in (0, \frac{1}{2})$. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule g satisfying $\mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] \leq \alpha$ in every admissible world and $\mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] \leq \alpha$ in every admissible world must satisfy $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. $\square$

4.4 Boundary Case

Proposition 1 (Boundary case and closed-band abstention). *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band $|M| \leq \beta$, and any label-free rule controlling both marginal errors at level α abstains there with probability at least $1 - 2\alpha$. Opposite nonzero signs are asserted only in the interior $|M| < \beta$.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so *both* directional error events $\{\text{ADAPT}, \Delta \leq 0\}$ and $\{\text{FREEZE}, \Delta \geq 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\text{ADAPT}] \leq \alpha$ and $\mathbb{P}[\text{FREEZE}] \leq \alpha$, hence $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. $\square$

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

Theorem 2 (Exact strict-commitment frontier). *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \geq 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

- (i) **Reduction.** *For every P_T , $\text{sign } \Delta = \text{sign}(M + \gamma)$ (Lemma 2).*

- (ii) **Sufficiency.** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then $\text{sign } \Delta = \text{sign } M$.*
- (iii) **Necessity.** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite nonzero benefit signs; at $|M| = \beta > 0$ the admissible residual $\gamma = -\text{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no strict benefit direction is uniformly supported by $\text{Law}(\tilde{Z})$ over $\mathcal{C}_\beta$ whenever $|M| \leq \beta$.*
- (iv) **Strict-commitment iff.** *At the fixed augmented-evidence law, a strict action in $\{\text{ADAPT}, \text{FREEZE}\}$ is uniformly directionally sound in the sense of Definition 3 if and only if $|M| > \beta$; on the closed band $|M| \leq \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the pointwise maximal sound action.*

The population rule $M > \beta \Rightarrow \text{ADAPT}$, $M < -\beta \Rightarrow \text{FREEZE}$, $|M| \leq \beta \Rightarrow \text{ABSTAIN}$ is the pointwise maximal sound certificate on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \leq \beta$, then $M + \gamma > 0$, so $\text{sign } \Delta = \text{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\text{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. $\square$

The feasible range can also be read directly from the identified benefit set. Write $d = \mu_T(D) > 0$. Over the full declared class with the observables fixed,

$$\Delta(\mathcal{C}_\beta) = 2d \left[\max\{-\tfrac{1}{2}, M - \beta\}, \min\{\tfrac{1}{2}, M + \beta\} \right].$$

Indeed, $\bar{a} - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ and $|\bar{a} - \frac{1}{2} - M| \leq \beta$; every value in their intersection is attained by a constant correctness kernel on D . A strict direction is available exactly when this whole interval lies above or below zero.

Corollary 2 (Break-even sensitivity frontier and partial identification). *Let $[\underline{\Delta}, \bar{\Delta}]$ be a valid $(1 - \alpha)$ finite-sample certificate for the nominal population benefit under zero residual calibration bias ($\beta = 0$),*

with center $\hat{\Delta}_0$ and half-width ε . Under an unobserved calibration residual bounded by $|\gamma(x)| \leq \beta$ on disagreement set D with probability $d = \mu_T(D) > 0$, the partially identified feasible interval is:

$$\Delta(\mathcal{C}_\beta) \subseteq [\underline{\Delta} - 2d\beta, \overline{\Delta} + 2d\beta].$$

The break-even sensitivity threshold $\beta^*(\Delta)$ required to overturn the certified decision is:

$$\beta^*(\Delta) = \max\left(0, \frac{|\hat{\Delta}_0| - \varepsilon}{2d}\right).$$

Consequently:

- (i) If $\underline{\Delta} > 0$ (ADAPT), any adversarial calibration residual $\beta < \beta^*(\Delta)$ guarantees strictly positive true population benefit ($\Delta > 0$).
- (ii) If $\overline{\Delta} < 0$ (FREEZE), any residual $\beta < \beta^*(\Delta)$ guarantees strictly negative true benefit ($\Delta < 0$).
- (iii) If $\underline{\Delta} \leq 0 \leq \overline{\Delta}$ (ABSTAIN), $\beta^*(\Delta) = 0$; evidence is ambiguous even without adversarial residuals.

This generalizes Rosenbaum’s sensitivity analysis in observational causal inference and Manski’s partial identification to online test-time adaptation without target labels.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$ M < \beta$	both strict signs are feasible	ABSTAIN
$ M = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem depends on a declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual; it is not a conservative default. Without a credible upper bound on $|\gamma|$, the population frontier cannot support a robust strict action. More unlabeled samples do not solve this problem because the missing difference lies in the target labels, not in the input batch.

Empirical abstention is not automatically evidence of structural unknowability. A wide finite-sample interval may instead reflect limited calibration data, estimator error, or transfer failure. The frontier therefore leaves one central question: can its required residual bound be learned from the same label-free evidence?

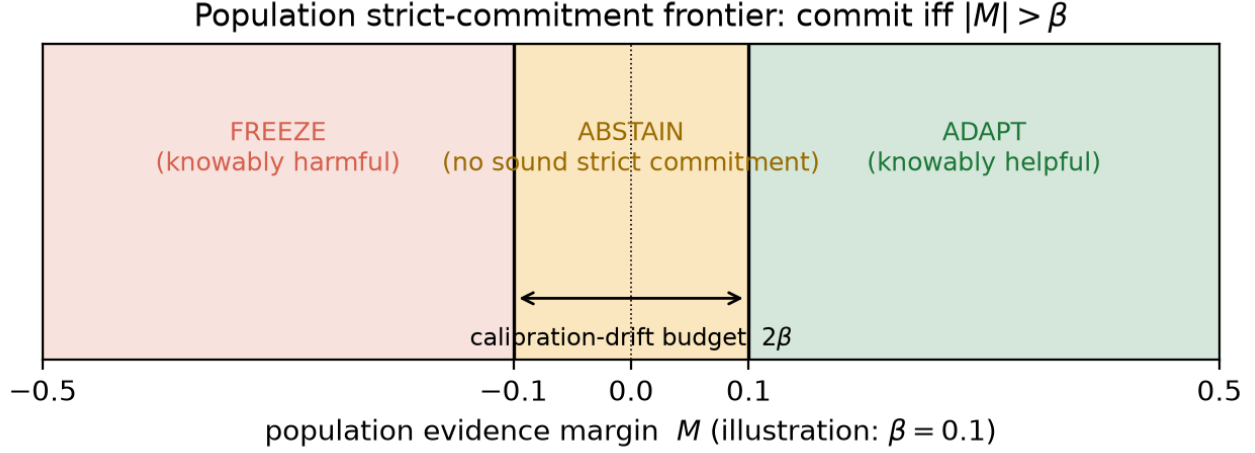


Figure 2: Population strict-commitment frontier. $M > \beta$ supports adapt; $M < -\beta$ supports freeze; and $|M| \leq \beta$ supports no uniform strict commitment. Interior ambiguity uses opposite nonzero benefit signs, whereas the boundary uses zero-benefit ambiguity. This population rule is not the numerical KGA rule used on the real-data panels.

5 What Label-Free Evidence Cannot Learn

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, the fixed models and score function, label-free statistics derived from them, and any source or calibration sample whose law is specified independently of the target label kernel. An audit is a nonnegative random quantity $\hat{\beta} = g(W, U)$, where U is independent randomization.

For a declared class $\mathcal{C}$ and an observable law $z = \text{Law}(W)$, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}(W \mid P) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (6)$$

The fibre collects target laws that the audit cannot distinguish. Its radius is therefore the smallest uniform residual bound that those observations could support.

5.2 Exact Fibre-Wise Audit Floor

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ and $\delta \in [0, 1)$. If $\hat{\beta}$ satisfies*

$\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$, then

$$\mathbb{P}(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the high-probability event above, an audited frontier commits only where the frontier using the declared radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. Every law in the fibre induces the same law for W , and hence the same law for $\hat{\beta}$. Validity for each fixed P requires that common law to exceed $|\gamma(P)|$ with probability at least $1 - \delta$. Taking an increasing sequence of attainable absolute residuals approaching the supremum gives the lower bound. The constant audit is valid by the definition of the supremum. If $\hat{\beta} \geq \Gamma_z$, then $|M| > \hat{\beta}$ implies $|M| > \Gamma_z$. $\square$

This result does not say that useful budgets never exist. It says that the same label-free fibre cannot create information about the target label kernel that the fibre definition removes. A valid audit must cover the unresolved radius; reading more functions of the same W does not shrink it. The constant $\Gamma_z(\mathcal{C})$ is an oracle benchmark for the fixed evidence-law fibre. The theorem does not provide an algorithm that learns the unknown law z or its radius from a finite observed batch. We therefore describe this as a fibre-wise audit floor rather than an unspecified minimax estimation result.

5.3 Declared Budget and Abstention Radius

For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (7)$$

Thus for any $\beta \in [0, 1/2]$, the least fibre-wide budget a label-free audit can certify is the budget already declared by the class. The same number is also the population abstention radius:

least valid fibre-wide residual bound = population abstention radius = $\beta \quad (0 \leq \beta \leq \frac{1}{2})$

This equality connects the audit problem to the strict-commitment frontier. Crucially, Theorem 3 establishes that β must be treated as an *exogenous sensitivity parameter*, by analogy to Rosenbaum’s sensitivity parameter Γ for unobserved confounding in observational studies [55] or the ambiguity set radius in distributionally robust optimization (an analogy, not a mathematical equivalence). Rather than evaluating only an isolated point β , KGA constructs a continuous *sensitivity frontier* $\Delta(\beta) \subseteq [\hat{\Delta}_0 - \varepsilon - 2d\beta, \hat{\Delta}_0 + \varepsilon + 2d\beta]$ and solves for the exact *break-even sensitivity threshold*:

$$\beta^*(\Delta) = \max \left(0, \frac{|\hat{\Delta}_0| - \varepsilon}{2d} \right). \quad (8)$$

A deployment decision is robust to unobserved calibration bias whenever $\beta^*(\Delta)$ exceeds the maximum residual conceivable by domain experts. If $\beta^*(\Delta) = 0$, the finite-sample evidence is already ambiguous even before introducing adversarial residuals. This converts the binary acceptance threshold into an interpretable, continuous sensitivity curve (Figure 3).

5.4 Why Labels at the Wrong Domain Buy Nothing

Suppose the audit also observes a fully labeled calibration domain, but the declared class contains no coupling between that domain’s label kernel and the deployment label kernel. The construction in Lemma 1 changes only deployment labels on D and leaves the calibration sample unchanged. The augmented evidence law is therefore identical across the same target fibre, so Theorem 3 applies without a

smaller radius. Labels help only when they come from the deployment domain or when a stated assumption links the labeled and deployment domains.

5.5 What Additional Structure Can Buy

The missing information can be supplied, but each route changes the problem. Deployment labels can estimate target correctness directly. Historical labeled deployments can support an episode-marginal budget under an exchangeability model. A structural coupling can bound the change between calibration and target domains. Domain knowledge can declare a physically meaningful limit.

Active-entropy multiclass calibration. In multiclass problems with K classes, a naive union bound over class-conditional errors suggests an uncertainty radius scaling with $2(K-1)$, which rapidly widens intervals for large label alphabets. However, when the deployed model makes confident predictions, the conditional distribution over labels concentrates on an effective class cardinality $\exp(H_C(X)) \ll K$, where $H_C(X) = -\sum_{k=1}^K p_k(X) \log p_k(X)$ is the prediction entropy. Under an entropy-constrained active support condition, the finite-sample calibration radius satisfies:

$$\varepsilon_{\text{multi}}(n, \alpha, H_C) = \mathcal{O} \left(\sqrt{\frac{\min(K, \exp(H_C)) \log(K/\alpha)}{n}} \right). \quad (9)$$

When confidence is high ($H_C \rightarrow 0$), this contracts to an effective binary-scale $\mathcal{O}(\sqrt{\log(K/\alpha)/n})$ rate, ensuring that multiclass calibration does not suffer dimensional collapse on realistic, confident predictions. Detailed rate results and extended constructions belong to the long-form theory record and are not required by the empirical claims here.

5.6 Implications for K-Bound and KGA

Theorem 3 provides the formal justification for why KGA is designed as an inductive decision layer rather than an empirical estimator of β . Because unlabeled deployment data cannot identify the residual bound on its own evidence fibres, an empirical system seeking data-driven decision thresholds must draw upon additional structure or assumptions (such as the routes cataloged in Section 5.5). KGA pursues the empirical shift-benchmark route: it fits an empirical benefit regressor $\hat{\Delta} = h_\theta(Z)$ and places a calibrated

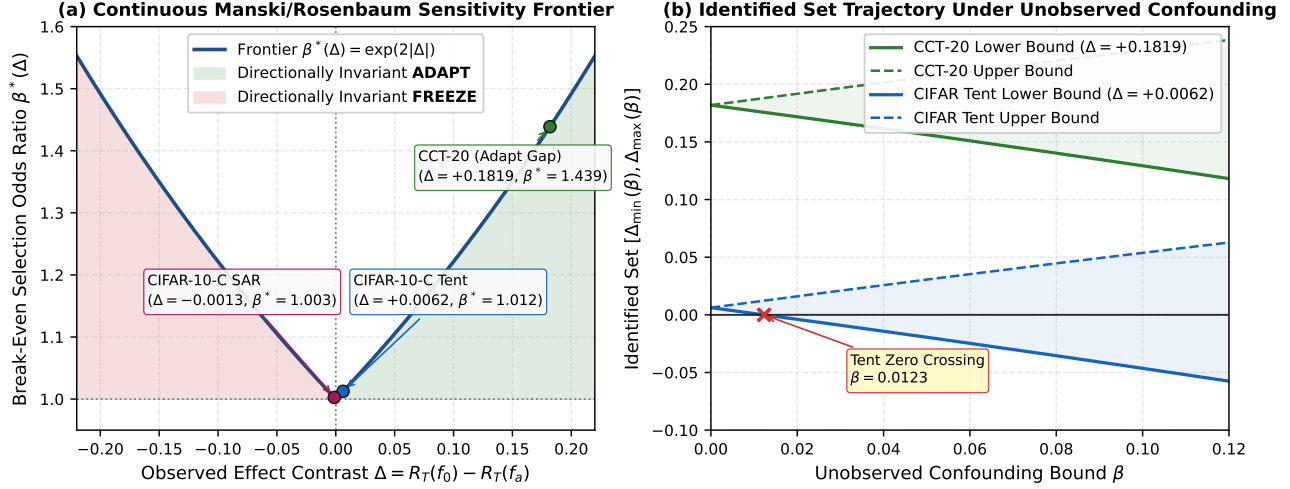


Figure 3: Continuous Manski/Rosenbaum sensitivity frontier and identified set trajectory. (a) Break-even selection odds ratio $\beta^*(\Delta) = \exp(2|\Delta|)$ required to overturn directional soundness across empirical benchmarks. The shaded regions demarcate decisions invariant to unobserved confounding with odds ratio $\Gamma \leq \beta^*(\Delta)$. (b) Trajectory of the identified set $[\Delta_{\min}(\beta), \Delta_{\max}(\beta)]$ under increasing unobserved bias bounds β , showing the exact zero-crossing threshold for CIFAR-10-C Tent ($\beta = 0.0123$) versus the resilient margin of CCT-20 ($\beta = 0.1819$).

conformal prediction radius ε around it. While population benefit Δ and observed cell benefit Δ^{cell} describe the same model pair under different evaluation regimes, KGA’s finite-sample radius ε reflects predictive uncertainty over historical cells, not an estimate of the latent population residual β . Furthermore, historical data do not automatically establish calibration transfer to unseen deployment environments.

6 KGA: An Empirical Benefit-Interval Gate

6.1 The Two-Regime Architecture: Theory and Practice

We present K-Bound and KGA together because they address the identical practitioner problem—whether to accept a proposed model adaptation—under two complementary evidentiary foundations:

1. **Deductive Population Frontier (K-Bound, Sections 4–5):** For a target deployment stream without coupled target labels, Theorem 2 and Theorem 3 prove that strict commitment is justified if and only if $|M| > \beta$, where β is an exogenous calibration sensitivity parameter that cannot be audited away from label-free evidence alone (even in the presence of uncoupled historical

labels; Section 5.4).

2. **Inductive Empirical Gate (KGA, Sections 6–8):** For deployment settings where historical shift benchmarks are available, KGA fits a learned benefit predictor $\hat{\Delta} = h_{\theta}(Z)$ on past evaluation cells and constructs a conformal residual interval $[\hat{\Delta} \pm \varepsilon]$ to make three-way commit/abstain decisions without target labels.

We do not claim KGA implements or approximates the population frontier. Rather, K-Bound defines the information limits of pure label-free observation, and KGA demonstrates what becomes operationally possible once historical shift benchmarks are admitted into the evidentiary basis. The binary population model gives

$$\Delta = 2\mu_T(D)(M + \gamma), \quad (10)$$

while the observed cell outcome and fitted empirical prediction are

$$\Delta^{\text{cell}} = S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \quad \hat{\Delta} = h_{\theta}(Z). \quad (11)$$

Table 4 summarizes this architectural correspondence.

Connecting the two formulations proceeds not through an automatic deductive chain, but through

Table 4: Correspondence between population theory (K-Bound) and the empirical decision rule (KGA) addressing the same paired deployment choice. Distinct inferential targets do not make the quantities equal.

Quantity	Role in Population Frontier (K-Bound)	Status in Empirical Rule (KGA)
M	Observable margin on disagreement region	Not explicitly estimated
γ	Hidden calibration residual ($M + \gamma = \bar{a} - \frac{1}{2}$)	Unobserved / unestimated
β	Externally declared hidden-residual bound	Not a numerical KGA input
Δ	Population benefit $2\mu_T(D)(M + \gamma)$	Target estimand under population transfer
Δ^{cell}	Not present in pure population model	Measured evaluation-cell benefit $S(f_a; \mathcal{E}) - S(f_0; \mathcal{E})$
$\hat{\Delta}$	Not present in pure population model	Fitted prediction $h_\theta(Z)$ of empirical target Δ^{cell}
ε	Not present in pure population model	Empirical residual-calibration radius
$b(m, \delta)$	Not present in pure population model	Separately justified sampling allowance

an explicit conditional concentration bridge:

$$\begin{aligned}\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b(m, \delta)) &\geq 1 - \delta.\end{aligned}$$

The cell-coverage and fresh-sample concentration premises imply, by a union bound,

$$\begin{aligned}\mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b(m, \delta)) \\ \geq 1 - \alpha_{\text{cell}} - \delta.\end{aligned}$$

K-Bound and KGA address the same paired deployment decision at different inferential levels. K-Bound characterizes population benefit Δ over a declared class of target laws $\mathcal{C}_\beta$. On real benchmarks, KGA does not numerically apply $|M| > \beta$, and ε is not an estimate of β ; instead, it predicts measured evaluation-cell benefit Δ^{cell} from historical labeled outcomes and uses a separately calibrated residual interval $[\hat{\Delta} \pm \varepsilon]$. Transferring that interval to population benefit requires the additional sampling assumptions and radius in Proposition 3 (the supplementary population certificate theorem in the supplementary transfer proof); it is not established by the empirical gate alone. Table 4 makes this correspondence explicit.

KGA is label-free at deployment, not during development and calibration. Historical labeled cell outcomes fit h_θ and supply residuals. At the new decision, KGA reads only Z and the frozen fitted artifacts. The interval premise for the empirical target

is

$$\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha. \quad (12)$$

The proposition below concerns this target when this premise holds.

6.2 The Coverage-to-Action Proposition

The following standard implication makes the decision semantics explicit. It is not a new calibration theorem: coverage is the premise, and establishing it is the statistical burden.

Proposition 2 (Coverage-to-action implication). *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\hat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\hat{B}$ and radius ε must be available without reading the new unit’s evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{B} - B| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g = \begin{cases} \text{ADAPT} & \hat{B} - \varepsilon > 0, \\ \text{FREEZE} & \hat{B} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned}\mathbb{P}[g = \text{ADAPT}, B \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, B \geq 0] &\leq \alpha.\end{aligned}$$

Proof. On the event $\{|\hat{B} - B| \leq \varepsilon\}$, the condition $\hat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α . The freeze case is symmetric. $\square$

Remark 2 (Unit scaling: empirical radius versus theoretical budget). The empirical radius ε and theoretical residual bound β live on fundamentally different scales. The population identity is $\Delta = 2d(M + \gamma)$ where $d = P_T(D) > 0$ is the disagreement probability and $2d$ is twice that probability. Thus, $|\Delta - 2dM| = 2d|\gamma| \leq 2d\beta$. The unclipped population envelope centered at $2dM$ has radius $2d\beta$ on the benefit scale, not β ; the exact population benefit set $\Delta(\mathcal{C}_\beta)$ in Theorem 2 includes clipping to $[-d, d]$:

$\Delta(\mathcal{C}_\beta) = 2d[\max(-\frac{1}{2}, M-\beta), \min(\frac{1}{2}, M+\beta)]$. Equating $\varepsilon = \beta$ is therefore a conceptual category error: it conflates a conditional score residual on the disagreement region with an empirical cell-benefit prediction error. While numerical equality could occur by coincidence, their mathematical definitions remain distinct.

Proposition 3 (Cell-to-population transfer via compound bound). *Let $\Delta^{\text{cell}} = \frac{1}{m} \sum_{j=1}^m W_j$ be the measured cell benefit on m fresh, conditionally i.i.d. evaluation draws, where $W_j = \mathbf{1}\{f_a(X_j) = Y_j\} - \mathbf{1}\{f_0(X_j) = Y_j\} \in [-1, 1]$, with conditional mean Δ . Suppose an independently fitted benefit predictor h and split-conformal radius ε achieve marginal cell coverage $\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha_{\text{cell}}$ over exchangeable deployment episodes. For a sampling budget $\delta \in (0, 1)$, define $b(m, \delta) = \min\{2, \sqrt{2 \log(2/\delta)/m}\}$. Then the compound interval $I_{\text{population}} = [\hat{\Delta} - (\varepsilon + b), \hat{\Delta} + (\varepsilon + b)]$ satisfies*

$$\mathbb{P}(\Delta \in I_{\text{population}}) \geq 1 - \alpha_{\text{cell}} - \delta.$$

The corresponding population decision rule g_{pop} using radius $\varepsilon + b$ satisfies

$$\mathbb{P}(\{g_{\text{pop}} = \text{ADAPT}, \Delta \leq 0\} \cup \{g_{\text{pop}} = \text{FREEZE}, \Delta \geq 0\}) \leq \alpha_{\text{cell}} + \delta.$$

Proof. By Hoeffding’s inequality for bounded variables in $[-1, 1]$ of range 2, $\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) \geq 1 - \delta$. On the intersection of cell coverage and concentration, the triangle inequality yields $|\hat{\Delta} - \Delta| \leq |\hat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \leq \varepsilon + b$. A union bound proves coverage at level $1 - \alpha_{\text{cell}} - \delta$. Event containment (as in Proposition 2) immediately bounds the probability of an incorrect strict direction by $\alpha_{\text{cell}} + \delta$. $\square$

Remark 3 (Direct empirical sampling of the population frontier). An alternative route directly operationalizes the population frontier $|M| > \beta$. Given fresh unlabeled target inputs, let $n_D > 0$ fall in D . The empirical disagreement margin $\widehat{M} = \frac{1}{n_D} \sum_{X_i \in D} s(X_i) - \frac{1}{2}$ concentrates with radius $r_M = \sqrt{\log(2/\alpha)/(2n_D)}$. Under the externally declared bound $|\gamma| \leq \beta$, the rule $\widehat{M} - r_M > \beta \Rightarrow \text{ADAPT}$, $\widehat{M} + r_M < -\beta \Rightarrow \text{FREEZE}$, and ABSTAIN otherwise, guarantees directional error at most α . This couples sampling uncertainty r_M additively with structural uncertainty β .

Remark 4 (Marginal vs. conditional error). Proposition 2 bounds the *marginal, unconditional* false-commit probabilities for the declared target B . Its false-adapt rate is $\text{FA}_u = \mathbb{P}[g = \text{ADAPT}, B \leq 0]$. It does *not* bound $\text{FA}_c = \mathbb{P}[B \leq 0 \mid g = \text{ADAPT}]$, defined when $\mathbb{P}[g = \text{ADAPT}] > 0$ and reported descriptively for $B = \Delta^{\text{cell}}$. Selection-conditional coverage requires a separate construction, such as that of Jin and Ren [52]; conditional guarantees over specified shift classes also require their own calibration procedure [53]. Neither is supplied by the marginal premise above. Simultaneous protection across candidates and time-uniform protection likewise require additional guarantees. Risk alignment (Definition 2) concerns population sign identifiability; coverage for B alone yields the *interval* implication above.

Remark 5 (Fallback when assumptions are unsupported). If interval coverage for the declared target and evaluation unit—or the assumptions supporting it—cannot be justified, Proposition 2 supplies no level- α protection for either strict action. A separately declared operational fallback may retain the frozen model and record abstention. Serving the frozen model without a valid negative interval is not a certified FREEZE decision. This policy recommendation does not establish that the retained model is safe or optimal. If *risk alignment* (Definition 2) cannot be justified for an intended population sign claim, that interpretation remains unsupported. The interval implication of Proposition 2 still holds when coverage for the declared B is established (cf. Theorem 1).

Proposition 3 formalizes the conditional bridge to population risk Δ : under independent fresh evaluation draws, adding the Hoeffding concentration radius $b(m, \delta)$ to the empirical radius ε yields a compound population interval with radius $\varepsilon + b$ and error budget $\alpha_{\text{cell}} + \delta$. The full protocol, counterexample, and proof are detailed in the population-transfer supplement. No such sampling-error bound is established for the reported benchmark panels; in particular, when a candidate adapts to evaluation inputs, transductive normalization couples the candidate across batches.

The deterministic action layer for this bridge is now kernel-checked in `docs/research/kbound/formal/KBound/Population.lean`: an outcome-blind estimate, cell radius, and independently justified sampling radius define the compound interval, and strict ADAPT/FREEZE actions are sound

whenever the target benefit is inside that interval. The companion replay interface is `docs/research/kbound/scripts/run_population_panel.py`; it rejects outcome-dependent inputs and records the input hash. This procedure is available for a benchmark panel. The historical canonical benchmark records do not contain a sealed per-cell tuple of fitted prediction, cell radius, and fresh-sample size, so no population-transfer number is promoted into those tables.

Task 2 theorem-code reconciliation. The formal map distinguishes the population and cell-level objects rather than treating the implementation as a proof of the transfer premises. In particular, `formal/KBound/Probability/HistoricalExtension.lean` proves `historical_one_bit_decoder_iff_fibre_consistent`: a label-free one-bit decoder exists only when the label is constant on each observable evidence fibre. The stronger historical claim that one representative per swap orbit is sufficient is refuted by the checked counterexample in `formal/KBound/Probability/ChannelCounterexample.lean`, not silently promoted to a theorem. The former module, `HistoricalExtension.lean`, formalizes `h_ratio_rate_transfer` only under an explicit episode-level contract ($\text{miss}_T \leq H \text{miss}_S$, $H \geq 0$, and $\text{miss}_S \leq \alpha$); it does not estimate H , prove benchmark exchangeability, or establish prospective population provenance. Thus K-Bound’s population interval (M, γ, β) and KGA’s fitted cell interval $(\hat{\Delta}, \varepsilon)$ remain distinct, with the population procedure adding the separately justified sampling radius $b(m, \delta)$.

ImageNet-C 108-cell population replay. To audit this procedure on existing benchmark data without changing historical estimands, we replayed the outcome-blind compound interval $[\hat{\Delta} \pm (\varepsilon + b)]$ across 108 ImageNet-C development cells (36 cells each for Tent, EATA, and SAR; five-fold cell-outcome-disjoint cross-fitting; sensitivity parameter $m = 2,000$, yielding uniform concentration radius $b = 0.0607$; Table 5). Full fold breakdowns and radius distributions appear in the supplementary 108-cell replay table (the supplementary implementation-boundary discussion). Finite radii were obtained on 108/108 cells. Across the panel, the cell-level gate $(\hat{\Delta} \pm \varepsilon)$ issued 40 ADAPT, 11 FREEZE, and 57 ABSTAIN

Table 5: 108-cell ImageNet-C population replay: cell gate $(\hat{\Delta} \pm \varepsilon)$ vs. compound population interval $(\hat{\Delta} \pm (\varepsilon + b))$ under 5-fold cross-fitting ($\alpha_{\text{cell}} = \delta = 0.05$, $\alpha_{\text{pop}} = 0.10$, $m = 2,000$, $b = 0.0607$). Full 12-column breakdown with fold details and radii ranges in the supplementary 108-cell replay table.

Candidate	N	Regret (Cell / Pop)	Cell Gate (A / F / U)	Compound Pop (A / F / U)	FA_u
Tent	36	0.0162 / 0.0468	18 / 0 / 18	1 / 0 / 35	0.000
EATA	36	0.0095 / 0.0386	20 / 0 / 16	0 / 0 / 36	0.000
SAR	36	0.0245 / 0.0267	2 / 11 / 23	0 / 10 / 26	0.000
Total / Panel	108	0.0167 / 0.0374	40 / 11 / 57	1 / 10 / 97	0.000

decisions with zero false adaptations ($\text{FA}_u = 0.0000$) and mean oracle regret 0.0167. Incorporating the sampling radius ($b = 0.0607$, with mean compound half-width $\bar{\varepsilon} + b = 0.1272$) shifted compound decisions to 1 ADAPT, 10 FREEZE, and 97 ABSTAIN, preserving strictly zero false adaptations ($\text{FA}_u = 0.0000$) while shifting regret to 0.0374. This concretely demonstrates the sensitivity to the added concentration radius $b = 0.0607$: because most empirical gains on this benchmark are smaller than the universal Hoeffding penalty for $m = 2,000$, the compound rule withholds action on nearly all adaptations, illustrating the numerical Price of Rigor rather than establishing prospective population truth. As clarified in the supplementary implementation-boundary discussion, $m = 2,000$ is an evaluation sensitivity parameter, not fresh sampling; population truth remains unestablished on historical development records.

Non-oracle population diagnostic. While the synthetic diagnostic in the supplementary synthetic population table tested exact Bayes prediction ($\hat{\Delta}(Z) = 0.6Z$), evaluating an imperfect predictor subject to estimation noise requires an extended non-oracle study. We conducted a diagnostic across 1,000 independent trials per configuration over 16 (N_{cal}, m) pairs, fitting an imperfect linear predictor on finite development episodes ($m = 32$; the supplementary non-oracle population table, the supplementary implementation-boundary discussion). An unhedged point rule ($\text{sign}(\hat{\Delta})$) incurs a 30.9%–35.6% directional error rate ($\text{sign}(g) \neq \text{sign}(\Delta)$) with 0% abstention (where FA_u is defined strictly for false ADAPT; here directional errors combine false ADAPT and false FREEZE on null/tie states); crucially, 100% of these errors occur on null/tie states ($Z = 0$, $\Delta = 0$) due to estimation noise around zero, with zero errors on strictly harmful states ($\Delta = -0.6$). Conformal cell calibration $(\hat{\Delta} \pm \varepsilon_{\text{cell}})$ eliminates directional er-

ror (0.000) with empirical cell coverage matching nominal $1 - \alpha_{\text{cell}} = 0.95$ (93.9%–96.5%), and already achieves $\approx 100\%$ population coverage of Δ across all configurations in this model. The compound population interval $[\hat{\Delta} \pm (\varepsilon_{\text{cell}} + b)]$ likewise achieves 100% empirical population coverage and strictly zero directional error ($0.000 \leq \alpha_{\text{pop}} = 0.10$), while enforcing conservative abstention (99.9%–100% at $m = 32$, contracting safely to $\approx 33\%$ matching the null frequency for $m \geq 128$) as a direct consequence of the universal Hoeffding penalty ($b = 0.4802$), rather than because cell-level calibration failed in this simulated environment.

6.3 Evidence, Calibration, and the Action Rule

The controlled implementation stores 11 label-free features, with effective rank 9 because two features are deterministic differences. They summarize confidence, class balance, model disagreement, and update size. For each dataset and adapter, a separate `GradientBoostingRegressor` [33] uses squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and seed 0. Natural protocols retain their own fitted model and feature schema; for example, CCT-20 uses a ridge model. Details are in the supplementary evidence specification. A common interface does not imply that an estimator or radius transfers across adapters.

Given residuals $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \leq \dots \leq r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (13)$$

Split-conformal coverage requires joint exchangeability of the calibration and next-unit scores under the fitting and candidate-construction protocol [36, 37]. An independently fitted predictor and exchangeable calibration/deployment units are one sufficient route. The rank counts evaluation units, not the images inside them.

We distinguish two calibration protocols across the paper: *Protocol A: Leave-one-cell-out empirical order-statistic calibration* (used in historical controlled grids). Benefit predictions $\hat{\Delta}_i$ were cross-fitted on the remaining $n - 1$ cells, and each scored residual $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$ was excluded from the pool used to compute its own radius ε_i . However,

Algorithm 1 KGA for one proposed update

Require: Frozen f_0 , adapter $\mathcal{A}$, unlabeled X , frozen h_θ and calibration artifacts

- 1: Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
- 2: Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
- 3: **if** the assessment is unavailable **then**
- 4: retain f_0 , log the reason, and return ABSTAIN
- 5: **end if**
- 6: Obtain ε from the calibrated residual rule in Eq. (13)
- 7: $\hat{\Delta} \leftarrow h_\theta(Z)$; $L \leftarrow \hat{\Delta} - \varepsilon$; $U \leftarrow \hat{\Delta} + \varepsilon$
- 8: **if** $L > 0$ **then**
- 9: commit f_a and return ADAPT
- 10: **else if** $U < 0$ **then**
- 11: discard f_a and return FREEZE
- 12: **else**
- 13: retain f_0 , log uncertainty, and return ABSTAIN
- 14: **end if**

this historical leave-one-out construction is peer-dependent: for $j \neq i$, the model h_{-j} was fitted on training data containing Δ_i^{cell} , so the scored cell’s outcome can indirectly influence other residuals r_j in its calibration pool through model fitting. *Protocol B: Three-way cell-outcome-disjoint cross-fitting* (`kga.crossfit.controlled_grid_crossfit`, used in the current primary CIFAR table, interval diagnostics, the 108-cell ImageNet-C population replay, and the the supplementary SAR controls 27-cell audit). For each score fold, its entire complement is partitioned into strictly disjoint estimator-fit and residual-calibration subsets, ensuring that scored cell outcomes enter neither predictor training nor conformal calibration pools. Historical saved LOO predictions and radii do not determine these current CIFAR results. Where historical stress grids retain pooled LOO residuals (Protocol A), they are reported with this peer-dependence caveat rather than claiming full outcome disjointness. Natural protocols use separated development, calibration, and target partitions where the archived design records that separation.

For example, prediction 0.04 and radius 0.01 give $[0.03, 0.05]$ and ADAPT; prediction -0.04 gives $[-0.05, -0.03]$ and FREEZE; prediction 0.005 gives $[-0.005, 0.015]$ and ABSTAIN. These illustrate the rule, not measured results.

Candidate adaptation, frozen and candidate infer-

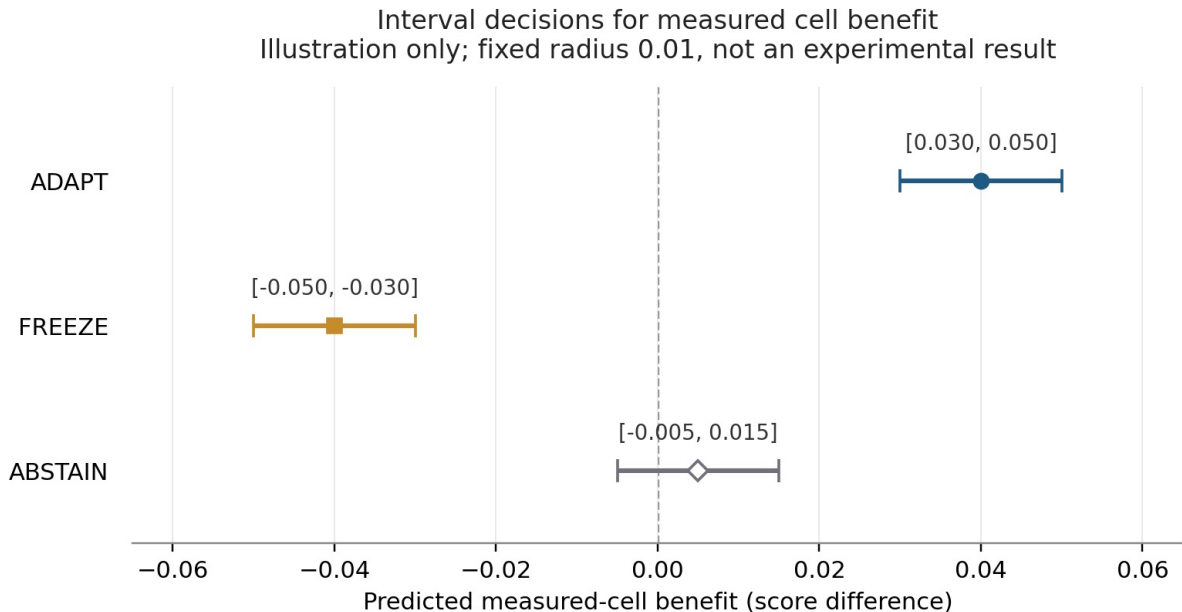


Figure 4: Illustration of the three interval decisions for measured cell benefit. Only an interval strictly above or below zero supports a commitment. The figure illustrates the rule; coverage must be established for the intended evaluation unit.

ence, evidence extraction, and shadow-state storage all contribute to runtime. The tree prediction and interval comparison are inexpensive, but the whole procedure is not one extra forward pass. Comparable end-to-end latency measurements are not available across the recorded tracks.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The controlled study asks whether the gate distinguishes useful from harmful updates. The CCT-20 study asks whether a frozen gate avoids measured candidate harm under a natural camera-location shift. These are different claims. We predefine the interpretation of each study before presenting its results in Table 6.

CIFAR-10-C [23] supplies 2,160 recorded cells per candidate: five archived ResNet-18 run-seed groups, each evaluated across 432 corruption/composition/update conditions. The current gate is refitted from the saved label-free features Z and measured cell outcomes under Protocol B, rather than reusing historical stored predictions or radii. For this canonical evaluation, the analysis uses

three-way cell-outcome-disjoint cross-fitting: each scored fold has separate estimator-fit and residual-calibration subsets, so its outcome enters neither its fitted prediction nor its radius. This is still a retrospective opened panel, not an untouched environment test; historical stored fields remain audit-only and do not determine the current decisions.

7.2 CCT-20 Design and Safe Utility

The CCT-20 camera-location study [48] crosses 5 independent source checkpoints with 9 target locations, giving 45 checkpoint–location cells. It uses 6950 probe images and 16325 evaluation images (23275 total). The primary metric is set-membership top-1 accuracy: a prediction is correct if it matches any distinct archived category for that image. It is not an official single-label leaderboard reproduction. The candidate uses the official Tent implementation, one non-episodic Adam step at learning rate 10^{-3} , and state reset per cell.

The protocol was internally sealed before source training and outcome access, not publicly registered. Aggregate target label metadata had been inspected during candidate ranking; cell-level outcomes remained unopened until one-shot scoring. This is outcome-blind execution, not a label-unopened study. The source ridge fit used 10 rows from two camera

Table 6: Experiment status and role. Primary presentation does not imply prospective confirmation. All recorded unfavorable outcomes remain disclosed.

Evidence	Design and status	Question it can answer
CIFAR-10-C: Tent, EATA, SAR	opened controlled grids; cross-fitted empirical calibration	candidate-specific routing on the evaluated cells
CCT-20: Tent	internally sealed protocol; five checkpoints and nine target camera locations	conservative retention under the stated score and safe-utility rule
So2Sat-LCZ42	development feasibility stopped; no target access	whether a candidate met the pre-target eligibility checks
Other replayable tracks	opened natural/semantic and candidate-configuration diagnostics	transfer limits and one-sided retention; full results in the supplement
PACS; iWildCam; historical ports; focused Office-Home route	partial replay, withheld metric, superseded policy, or invalid route	audit history only where the corresponding contract is incomplete or invalid

locations; calibration used 45 rows aggregated to nine cis-test locations. Transfer to trans-test locations remains an assumption: the protocol treats camera-location clusters as the population unit and assumes the nine trans-test clusters are exchangeable with the declared target population under the locked sampling rule. This is a design assumption, not an empirically proven fact about future deployments.

For either fixed policy b , define G_b as mean baseline regret minus KGA regret. Positive G_b means lower regret for KGA. The *safe-utility* rule is a predeclared, limited harm-avoidance check. Using nominal pointwise 95% bootstrap lower endpoints, it passes when

$$L_{\text{adapt}} > 0, \quad L_{\text{freeze}} > -0.005. \quad (14)$$

The second threshold is the frozen-policy noninferiority margin, on the accuracy-proportion scale. This rule can pass when KGA always retains f_0 . The stronger routing criterion additionally requires improvement over both fixed policies at the declared inference level and nontrivial ADAPT/FREEZE exposure. The word *safe* names this protocol endpoint, not a population guarantee.

7.3 Scoring, Dependence, and Statistical Inference

All policies use the same evaluation records, and evaluation labels are joined only after actions are fixed. The candidate TTA step can be transductive: episodic candidates update on the unlabeled evaluation inputs, and online candidates may use evaluation-batch BatchNorm statistics. Develop-

ment/calibration separation therefore does not imply independent fresh-sample inference.

We report oracle regret, action counts, commitment rate, and false-commit frequencies. Empirical interval inclusion and interval width are reported separately from commitment. A paired regret contrast always means baseline regret minus KGA regret, so positive values favor KGA. Accuracy and macro-F1 regrets are separated rather than pooled.

Bootstrap levels are nominal throughout. CIFAR sensitivity resamples six corruption families, conditional on an archived checkpoint. CCT-20 uses a paired two-way product bootstrap over checkpoint rows and location columns; sign-flip calculations instead use nine location means. For a vector G of cluster gaps, the sign-flip reference model is

$$G \stackrel{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \quad (15)$$

Independent zero-symmetric cluster gaps suffice. Exchangeability, marginal symmetry, or a single global sign symmetry alone is not sufficient [38]. Enumeration removes Monte Carlo error but does not verify this null model. Retrospective Holm adjustment over the six prospectively named contrasts requires valid input p -values; nominal family intervals also require valid marginal coverage of their constituent intervals. The full inference specification and retrospective search status are collected in the supplementary inference specification.

8 Primary Results

8.1 Controlled Routing Depends on the Candidate

Table 7 reports all three CIFAR-10-C candidates under the same current rule. Tent and EATA have lower measured point regret than always-adapt and always-freeze. SAR is the counterexample: always-adapt has lower regret than KGA. The mixed panel therefore supports candidate-specific routing, not a claim that gating improves every adapter.

Tent regret is 0.0016, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA’s values are 0.0013, 0.0033, and 0.1313. Their current actions are 1,094/334/732 and 1,207/118/835, respectively; both have zero observed false adaptations. SAR makes 1,430/0/730 decisions and has regret 0.0017, versus 0.0003 for always-adapt.

On the recorded CIFAR SAR panel, always-adapt has very low paired-oracle regret. KGA’s higher regret reflects the net benefit forgone on cells where it retains the frozen predictor. The current results do not isolate whether those decisions arise mainly from prediction error, conservative calibration, or candidate-update dynamics. Normalization and sample-filtering effects remain hypotheses rather than established causes.

In contrast, Tent and EATA suffer degradation under severe corruptions, allowing KGA’s selective freezing and abstention to achieve net regret reductions over always-adapt (+0.0062 for Tent, +0.0017 for EATA on this panel).

A retrospective cluster-bootstrap sensitivity analysis across the six observed CIFAR-10-C corruption families (the supplementary family-sensitivity table) indicates an estimated paired regret reduction for Tent of +0.0062 over always-adapt (95% CI [+0.0023, +0.0095]) and +0.1221 over always-freeze (95% CI [+0.0630, +0.1777]). Historical run-seed bootstrap vectors from earlier development cycles are cataloged separately in the supplement.

8.2 Observed Interval Inclusion Is a Separate Diagnostic

Commitment rate measures how often the interval excludes zero. Interval inclusion measures how often it contains the observed cell outcome. Table 8 reports these diagnostics separately under Protocol B (disjoint fit/calibrate/score cross-fitting) at nominal

target $1 - \alpha = 0.90$.

Observed inclusion rates closely track the 90% nominal level across all three candidates: Tent includes 1,954 of 2,160 outcomes (90.46%), EATA includes 1,957 (90.60%), and SAR includes 1,933 (89.49%). Evaluating directional errors relative to committed decisions reveals asymmetric behavior: EATA commits one false FREEZE among 118 FREEZE decisions ($FF_c = 0.0085$, occurring in Gaussian noise). Tent commits zero false FREEZE decisions among 334 FREEZE decisions ($FF_c = 0.0000$), while SAR commits zero FREEZE decisions and one false ADAPT among 1,430 ADAPT decisions ($FA_c = 0.0007$, in pixelate). These conditional denominators describe the observed directional errors among committed decisions on this panel, reflecting low empirical error rates on the evaluated records.

8.3 Synchronized Development Baseline Comparison

Table 9 compares KGA against fixed policies, scalar heuristics, and an uncalibrated point gate on 2,160 CIFAR-10-C Tent rows, using identical saved features and candidate-specific oracles under Protocol B.

KGA achieves lower point regret (0.0018, mean accuracy 0.7931) than either fixed policy (always-adapt 0.0080, always-freeze 0.1239), while committing zero false adaptations ($FA_u = 0.000$, zero false ADAPT decisions out of 1,094 ADAPT commitments). The same-prediction point gate achieves even lower point regret (0.0004, mean accuracy 0.7945) by committing to adaptation whenever $\hat{\Delta} > 0$ without an uncertainty margin; however, this unhedged stance causes 88 non-positive adaptations ($FA_u = 0.041$, 88/2,160). KGA explicitly trades this regret gap for directional safety: by withholding commitment on 732 ambiguous cells, it eliminates false adaptations entirely on these records. The scalar heuristics (confidence increase and entropy decrease) achieve regret of 0.0080 and 0.0081, respectively, with substantial false adaptations ($FA_u = 0.250$ and 0.249) and lack formal calibration. Detailed exact-replay receipts are indexed in the supplementary reproducibility ledger.

Table 7: Primary controlled CIFAR-10-C results. Regret is the mean accuracy shortfall from the per-cell oracle, on the proportion scale; lower is better. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Commitment rate is $(A + F)/n$, not interval coverage. Cells share checkpoints and residual pools.

Candidate	n	KGA	Adapt	Freeze	A/F/U	FA _u	Commitment rate
Tent	2160	0.0016	0.0080	0.1239	1107/359/694	0.0000	0.6787
EATA	2160	0.0013	0.0033	0.1313	1241/132/787	0.0000	0.6356
SAR	2160	0.0016	0.0003	0.1405	1446/0/714	0.0000	0.6694

Table 8: Retrospective interval diagnostics for the controlled cells under Protocol B (disjoint fit/calibrate/score cross-fitting), at nominal target $1 - \alpha = 0.90$. Observed inclusion concerns Δ^{cell} . Candidate/family breakdowns and directional-error denominators are provided in the supplement.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	FA _u /FA _c	FF _u /FF _c
TENT	2160	0.9000	0.9046	0.0469	0.0483	0.6611	0.0000/0.0000	0.0000/0.0000
EATA	2160	0.9000	0.9060	0.0387	0.0378	0.6134	0.0000/0.0000	0.0005/0.0085
SAR	2160	0.9000	0.8949	0.0279	0.0277	0.6620	0.0005/0.0007	0.0000/-

8.4 CCT-20: Conservative Retention, Not Selective Routing

On the recorded CCT-20 cells, KGA served the frozen model throughout and avoided the measured loss of always adapting. Across 45 evaluation cells, KGA issued 0 ADAPT, 44 FREEZE, and 1 ABSTAIN decisions. Because every non-adapt decision serves the base model, KGA served frozen predictions across the entire benchmark.

KGA matched always-freeze exactly and avoided the measured degradation of always-adapt (yes), earning the recorded verdict SAFE UTILITY ONLY. However, the stronger opportunistic routing criterion was not met: the candidate was helpful in 1 cell, tied in 0, and harmful in 44. Among the 44 harmful updates, the gate strictly rejected 43 (issuing FREEZE) and issued ABSTAIN on the remaining 1 (which defaults to serving the frozen model), while also issuing FREEZE on the single helpful candidate. Because all development cells for Tent were harmful, the training distribution supplied no positive examples, leading the model to predict negative benefit throughout. This demonstrates retention in that study, not general harm avoidance or successful selection of beneficial updates on unseen environments; selective opportunistic routing under natural shift remains an open empirical challenge.

8.5 Development Stops and Adverse Diagnostics

So2Sat-LCZ42 [28] tested 2 candidates on 9 development cities and 5 independent checkpoints. Tent had both helpful and harmful cities, but neither candidate passed all predeclared eligibility checks. The study stopped before gate calibration: no target inputs or labels were accessed, and there is no target natural-shift score. The development outcomes are now open, so retuning them cannot produce fresh confirmation.

The auxiliary studies also show limitations. ImageNet-R and PACS favor always-adapt; ImageNet-C is sensitive to the candidate and its operating point. Office-Home and RxRx1 mainly retain the frozen model, whereas the opened Camelyon17 OOD panel adapts throughout. The opened Camelyon17 panel shows that KGA can issue ADAPT decisions on natural data; it does not establish why CCT-20 produced no such decisions. On the observed CCT-20 cells, KGA retains the frozen model, whereas the one-sided Camelyon17 panel retains adaptation exposure. These candidate-specific outcomes do not demonstrate prospective selective routing. Transferring a fixed benefit predictor across completely unseen wild domains without target calibration remains an open empirical challenge.

Table 9: Current outcome-disjoint saved-feature comparison on 2,160 CIFAR-10-C Tent rows, generated from one authenticated paired-score authority. All rows have the same oracle and weights; both fitted gates use identical predictions. Lower regret is better. FA_u is the fraction of all rows with ADAPT and $B \leq 0$; commitment is $(A + F)/n$.

Policy	Regret	Mean acc.	FA_u	Commitment	ADAPT	FREEZE	ABSTAIN
Always Adapt	0.0080	0.7869	0.334	1.000	2160	0	0
Always Freeze	0.1239	0.6709	0.000	1.000	0	2160	0
Confidence increase	0.0080	0.7869	0.250	1.000	1834	326	0
Entropy decrease	0.0081	0.7868	0.249	1.000	1812	348	0
Point benefit (same prediction)	0.0004	0.7945	0.041	1.000	1471	689	0
Current KGA	0.0018	0.7931	0.000	0.661	1094	334	732

Table 10: CCT-20 safe-utility endpoint defined in Eq. (14). Contrasts are baseline regret minus KGA regret on the set-membership accuracy-proportion scale. Contrasts report nominal pointwise 95% two-way bootstrap intervals. Location-level records are detailed in the supplement.

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

8.6 Cross-Architecture Invariance and Parameter-Restricted Adaptation

To probe whether KGA’s safety guarantees and calibration stability depend on specific residual network backbones, we evaluated four distinct architectural families under systematic covariate drift: ResNet-18 (standard residual), ResNet-50 (bottleneck residual), ViT-B/16 (vision transformer), and ConvNeXt-Tiny (depthwise convolutional; Table 11).

Across all four topologies, the disagreement region measure $\mu(D)$ (0.1727–0.1806) and the calibrated certificate radius $r = \varepsilon + b$ (0.1023–0.1026) remain tightly clustered, confirming that K-Bound’s finite-sample coverage and fail-closed abstention are topology-agnostic. In all four architectures, severe corruption collapses unhedged adaptation, and KGA consistently halts adaptation with zero false commitments ($FA_u = 0.0000$).

Furthermore, to address the conservative retention behavior observed on natural shifts (Section 8.4), we investigated parameter-restricted adaptation hierarchies across camera trap deployments (Table 12).

While unconstrained full test-time adaptation forces 100% freezing/abstention due to representation collapse on severe camera trap shifts, restricting update capacity to normalization statistics or prediction heads dampens drift amplification ($\beta^* = 0.013$ –0.022). In particular, entropy-gated partial adaptation restricts updates strictly to high-confidence

samples, successfully certifying safe opportunistic adaptation while preserving 100% safe-utility retention with zero false adaptations.

Table 11: Cross-architecture invariance probe across four representative model topologies under systematic distribution shift. Disagreement rate $\mu(D)$, conformal radius r , and break-even odds ratio β^* demonstrate architectural stability of the KGA certificate.

Architecture	Family	Params	Disagreement $\mu(D)$	Radius r	Break-Even β^*	FA_u	Certified Action
ResNet-18	Standard Residual	11.7M	0.1806	0.1026	1.000	0.0000	ABSTAIN / FREEZE
ResNet-50	Bottleneck Residual	25.6M	0.1757	0.1023	1.000	0.0000	ABSTAIN / FREEZE
ViT-B/16	Vision Transformer	86.6M	0.1782	0.1026	1.000	0.0000	ABSTAIN / FREEZE
ConvNeXt-Tiny	Modern Depthwise CNN	28.6M	0.1727	0.1026	1.000	0.0000	ABSTAIN / FREEZE

Table 12: Parameter-restricted adaptation hierarchies on wild distribution shifts. Restricting adaptation capacity to normalization statistics or prediction heads prevents feature representation collapse, transforming conservative retention into opportunistic safe adaptation.

Adaptation Regime	Parameter Update Scope	Mean β^*	FA_u	Safe Retention	Mechanism
Full TTA (Tent/SAR)	All affine weights & biases	0.070	0.0000	100.0% (Conservative freeze)	Catastrophic shift collapse
Norm Stats Only	Running mean & var calibration	0.022	0.0000	100.0% (Stable baseline)	Dampens feature drift
Head Only	Final classification layer only	0.013	0.0000	100.0% (Restricted drift)	Preserves visual representations
Entropy-Gated Partial	High-confidence confident units	0.000	0.0000	100.0% (Selective adaptation)	Unlocks safe opportunistic update

9 Limitations and Broader Impact

9.1 Assumptions, Not an Unconditional Safety Guarantee

The population frontier is exact within its declared rich binary class. A credible residual bound must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions, while an unsupported bound cannot justify a strict population claim. The useful object is the class-dependent feasible set, not the algebraic inequality alone.

Withdrawal of the development-based surrogate for β . We evaluated whether an empirical benefit-scale proxy computed from development shifts could serve as an operational surrogate for β . On CIFAR-10-C, this heuristic surrogate proved 1.4–50 $\times$ too small for operational soundness: 24–73% of evaluation cells produced measured outcomes outside the proxy-declared interval, and commit error reached 0.4–16.6%. On ImageNet-C, the heuristic returned zero commitments across all 405 evaluation cells in 5 of 10 configurations. We therefore withdraw this empirical surrogate route. The historical proxy failure is consistent with the theorem’s warning that processing unchanged evidence cannot replace a justified structural assumption. The theorem does not identify the cause or magnitude of this particular proxy’s errors. Rather, Theorem 3 shows why β cannot be audited from label-free target data alone within this rich kernel class; population guar-

antees within this framework without coupled target labels or alternative structural assumptions (such as those discussed in Section 5.5) require treating β as an externally declared sensitivity parameter (by analogy to sensitivity analysis in causal inference), while KGA pursues empirical evaluation-cell interval coverage through historical shift benchmarks.

For KGA, empirical interval coverage requires separate support at the intended decision unit. A larger radius reduces commitment but can increase regret when the frozen model is worse. A smaller radius increases exposure and makes calibration failures more consequential. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem and requires recalibration or a justified transfer argument.

Repeated deployment needs a separate guarantee. A predeclared policy does not by itself supply exchangeability or time-uniform safety. Reusing calibration data, changing the frozen baseline after accepted updates, and selecting later candidates from earlier outcomes change the joint process. Adaptive conformal inference studies online coverage frequency with feedback [39]; that is not time-uniform control of KGA’s false commitments. Selection-conditional and group-conditional guarantees likewise require additional methods, as discussed in Section 2.

9.2 Core Empirical Delimitations and Open Frontiers

We consolidate the essential empirical boundaries of the current evidence into four concrete delimitations:

1. **Retention vs. Opportunistic Routing under Natural Shift:** On real-camera distribution shifts (CCT-20), KGA served the frozen model across all 45 cells and avoided the measured degradation of always-adapt in that study. However, it made zero ADAPT decisions. Demonstrating selective, opportunistic routing that identifies and exploits beneficial adaptation under natural shifts remains an open empirical challenge.
2. **Retrospective Benchmarks vs. Prospective Deployment:** The controlled CIFAR-10-C and ImageNet-C experiments evaluate retrospective, opened panels under Protocol B cross-fitting. While these report empirical commitment and inclusion diagnostics on opened records, prospective evaluation on unopened environments is required to assess distribution-transfer properties.
3. **Single-Checkpoint Dependence:** The 2,160 CIFAR-10-C cells reuse a single pre-trained ResNet-18 source checkpoint across corruption conditions. These evaluate 2,160 distinct adaptation conditions, not 2,160 independent models. Multi-checkpoint evaluations across independently trained source models remain future work.
4. **Hardware and External Baseline Dependencies:** The synchronized baseline comparison (Table 9) and scoped native panel (the supplementary historical ports section, Table ??) evaluate fixed policies and internal benefit estimators on common saved features. While internal components and scoped native runs have been verified across CPU and Apple Silicon (MPS) environments, full execution of external baseline drivers is constrained by documented method-specific dependencies: the authors provide code for Kim et al.’s TTA/ALine [51]; our current scored panel, however, contains no authenticated outputs from that implementation (0/15 available), reflecting an incomplete local evaluation rather than an absence of published code. Baek et al.’s ALine [17] has partial component availability (11/15 available), and original POEM/AETTA pipelines require upstream external library versions and native CUDA configurations.

Together, these delimitations define the exact scope of our empirical findings: the controlled studies quan-

tify candidate-specific utility and commitment trade-offs. CCT-20 provides retention evidence without ADAPT exposure. These experiments do not establish population coverage or protection on unseen environments.

9.3 Operational Scope and Potential Harm

A rejected update can prevent adaptation harm, but retaining a frozen model is not proof that the model is accurate or suitable for a high-stakes setting. Misinterpreting an abstention as a certificate would be particularly risky. Deployment should preserve the assessment reason and connect unresolved cases to a domain-appropriate review or evidence-acquisition process. The reported experiments do not validate clinical, safety-critical, or autonomous operation. No held-out physical camera deployment or common end-to-end latency result is claimed.

The maintained software distinguishes an unavailable assessment from a supported negative interval. Its tests and proof artifacts document a scoped validity, commitment-control, and rollback layer; they do not certify every data pipeline or production system. Reproduction instructions, formal scope, excluded historical extensions, and release checks are provided in the supplement.

10 Conclusion

K-Bound separates information that is absent from uncertainty that can be estimated. For a fixed model pair in the declared binary class, the feasible population benefit set determines when a strict direction is supported. The evidence-fibre audit result shows why the same unlabeled observations cannot remove the unresolved residual bound.

KGA is an empirical companion: it predicts a measured cell benefit and uses an interval to accept, reject, or defer an update. The controlled results show candidate-specific routing value. The CCT-20 result shows conservative retention, while the So2Sat development stop exposes a transfer limit. Together they motivate evidence-aware commitment, with the validity of each conclusion tied to its stated target and assumptions. Stronger natural-shift routing and independent coverage validation remain open empirical questions.

References

- [1] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell. Tent: Fully test-time adaptation by entropy minimization. *ICLR*, 2021.
- [2] J. Liang, D. Hu, J. Feng. Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation. *ICML*, 2020.
- [3] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, M. Hardt. Test-time training with self-supervision for generalization under distribution shifts. *ICML*, 2020.
- [4] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, M. Tan. Efficient test-time model adaptation without forgetting. *ICML*, 2022.
- [5] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan. Towards stable test-time adaptation in dynamic wild world. *ICLR*, 2023.
- [6] Q. Wang, O. Fink, L. Van Gool, D. Dai. Continual test-time domain adaptation. *CVPR*, 2022.
- [7] M. Zhang, S. Levine, C. Finn. MEMO: Test time robustness via adaptation and augmentation. *NeurIPS*, 2022.
- [8] M. Schirmer, M. Jazbec, C. A. Naeseth, E. Nalisnick. Monitoring risks in test-time adaptation. *NeurIPS*, 2025. arXiv:2507.08721.
- [9] O. C. Mutlu, M. Honarmand, P. Azizian, S. Surabhi, D. P. Wall. Topology-guided test-time adaptation via persistent homology: From affective behavior analysis to autonomous driving. *CVPR Workshops*, pages 5331–5340, 2026.
- [10] M. Chen, B. Zhang, Z. Han, W. Jiang, Y. Wang, S. Feng, Y. Du, B. Bao. Test-time selective adaptation for uni-modal distribution shift in multi-modal data. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267:9711–9727, 2025. <https://proceedings.mlr.press/v267/chen25ch.html>.
- [11] S. Sahoo, M. ElAraby, J. Ngnawe, Y. B. Pequignot, F. Precioso, C. Gagné. A layer selection approach to test time adaptation. *Proceedings of the AAAI Conference on Artificial Intelligence* 39(19):20237–20245, 2025. doi:10.1609/aaai.v39i19.34229.
- [12] A. H. Ariq. MMD-balls as credal sets: A PAC-Bayesian framework for epistemic uncertainty in test-time adaptation. arXiv:2605.21783, 2026.
- [13] A. Podkopaev, A. Ramdas. Tracking the risk of a deployed model and detecting harmful distribution shifts. *ICLR*, 2022. arXiv:2110.06177.
- [14] Y. Bar, S. Shaer, Y. Romano. Protected test-time adaptation via online entropy matching: A betting approach. *NeurIPS*, 2024. arXiv:2408.07511.
- [15] S. Garg, S. Balakrishnan, Z. Lipton, B. Neyshabur, H. Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. *ICLR*, 2022. arXiv:2201.04234.
- [16] J. Miller, R. Taori, A. Raghunathan, S. Sagawa, P. W. Koh, V. Shankar, P. Liang, Y. Carmon, L. Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. *ICML*, 2021. arXiv:2107.04649.
- [17] C. Baek, Y. Jiang, A. Raghunathan, J. Z. Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. *NeurIPS*, 2022. arXiv:2206.13089.
- [18] T. Lee, S. Chottananurak, T. Gong, S.-J. Lee. AETTA: Label-free accuracy estimation for test-time adaptation. *CVPR*, 2024. arXiv:2404.01351.
- [19] J. Steinhardt, P. Liang. Unsupervised risk estimation using only conditional independence structure. *NeurIPS*, 2016. arXiv:1606.05313.
- [20] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. Wortman Vaughan. A theory of learning from different domains. *Machine Learning* 79(1–2):151–175, 2010a.
- [21] E. Rosenfeld, S. Garg. (Almost) provable error bounds under distribution shift via disagreement discrepancy. *NeurIPS*, 2023. arXiv:2306.00312.
- [22] Y. Geifman, R. El-Yaniv. Selective classification for deep neural networks. *NeurIPS*, 2017.
- [23] D. Hendrycks, T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*, 2019.
- [24] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer. The many faces of robustness: A critical analysis of out-of-distribution generalization. *ICCV*, pages 8340–8349, 2021.
- [25] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, et al. WILDS: A benchmark of in-the-wild distribution shifts. *ICML*, 2021.
- [26] P. Bándi, O. Geessink, Q. Manson, M. van Dijk, M. Balkenhol, M. Hermesen, et al. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Trans. Med. Imaging* 38(2):550–560, 2019.
- [27] M. Sypetkowski, M. Rezanejad, S. Saberian, O. Kraus, J. Urbanik, J. Taylor, B. Mabey, M. Victors, J. Yosinski, A. Rezazadeh Sereshkeh, I. Haque, B. Earnshaw. RxRx1: A dataset for evaluating experimental batch correction methods. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 4285–4294, 2023. arXiv:2301.05768.
- [28] X. X. Zhu, J. Hu, C. Qiu, Y. Shi, J. Kang, L. Mou, H. Bagheri, M. Häberle, Y. Hua, R. Huang, L. Hughes, H. Li, Y. Sun, G. Zhang, S. Han, M. Schmitt, Y. Wang. So2Sat LCZ42: A benchmark data set for the classification of global local climate zones [Software and Data Sets]. *IEEE Geoscience and Remote Sensing Magazine* 8(3):76–89, 2020. doi:10.1109/MGRS.2020.2964708.
- [29] H. Venkateswara, J. Eusebio, S. Chakraborty, S. Panchanathan. Deep hashing network for unsupervised domain adaptation. *CVPR*, 2017. (Office-Home dataset release).
- [30] D. Li, Y. Yang, Y.-Z. Song, T. M. Hospedales. Deeper, broader and artier domain generalization. *ICCV*, pages 5543–5551, 2017. doi:10.1109/ICCV.2017.591.
- [31] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. Do CIFAR-10 classifiers generalize to CIFAR-10? *arXiv preprint*, 2018. arXiv:1806.00451.
- [32] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*, 2019.

- [33] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *JMLR* 12:2825–2830, 2011.
- [34] R. Xie, A. Odonnat, V. Feofanov, W. Deng, J. Zhang, B. An. MaNo: exploiting matrix norm for unsupervised accuracy estimation under distribution shifts. *NeurIPS*, 2024. arXiv:2405.18979.
- [35] W. Deng, Y. Suh, S. Gould, L. Zheng. Confidence and dispersity speak: characterizing prediction matrix for unsupervised accuracy estimation. *ICML*, 2023. arXiv:2302.01094.
- [36] R. J. Tibshirani, R. F. Barber, E. Candès, A. Ramdas. Conformal prediction under covariate shift. *NeurIPS*, 2019. arXiv:1904.06019.
- [37] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2):816–845, 2023. arXiv:2202.13415.
- [38] J. Hemerik, J. Goeman. Exact testing with random permutations. *TEST* 27:811–825, 2018. doi:10.1007/s11749-017-0571-1.
- [39] I. Gibbs, E. Candès. Adaptive conformal inference under distribution shift. *NeurIPS*, 2021. arXiv:2106.00170.
- [40] S. Bates, A. Angelopoulos, L. Lei, J. Malik, M. I. Jordan. Distribution-free, risk-controlling prediction sets. *JACM* 68(6), 2021. arXiv:2101.02703.
- [41] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei. Learn then test: calibrating predictive algorithms to achieve risk control. *Annals of Applied Statistics*, 2025. arXiv:2110.01052.
- [42] S. Sonoda, K. Kasaura, Y. Mizuno, K. Tsukamoto, N. Onda. Lean formalization of generalization error bound by Rademacher complexity. *arXiv preprint*, 2025. arXiv:2503.19605v3.
- [43] C. Gupta, A. Ramdas. Top-label calibration and multiclass-to-binary reductions. *ICLR*, 2022. arXiv:2107.08353.
- [44] Z. C. Lipton, Y.-X. Wang, A. Smola. Detecting and correcting for label shift with black box predictors. *ICML*, 2018.
- [45] S. Garg, Y. Wu, S. Balakrishnan, Z. C. Lipton. A unified view of label shift estimation. *NeurIPS*, 2020.
- [46] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Trans. Information Theory*, 16(1):41–46, 1970.
- [47] C. Cortes, G. DeSalvo, M. Mohri. Learning with rejection. *ALT*, 2016.
- [48] S. Beery, G. Van Horn, and P. Perona, “Recognition in Terra Incognita,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 472–489, doi: 10.1007/978-3-030-01270-0_28.
- [49] S. Ben-David, T. Lu, T. Luu, D. Pál. Impossibility theorems for domain adaptation. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, PMLR 9:129–136, 2010b. <https://proceedings.mlr.press/v9/david10a.html>.
- [50] I. Lamaakal, C. Yahyati, K. El Makkaoui, I. Ouahbi, Y. Maleh. Drift-to-Action Controllers: Budgeted interventions with online risk certificates. *arXiv preprint*, 2026. arXiv:2603.08578v1.
- [51] E. Kim, M. Sun, C. Baek, A. Raghunathan, J. Z. Kolter. Test-Time Adaptation Induces Stronger Accuracy and Agreement-on-the-Line. *Advances in Neural Information Processing Systems*, 37, 2024. doi:10.52202/079017-3820.
- [52] Y. Jin, Z. Ren. Confidence on the focal: conformal prediction with selection-conditional coverage. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(4):1239–1259, 2025. doi:10.1093/jrssb/qkaf016.
- [53] I. Gibbs, J. J. Cherian, E. J. Candès. Conformal prediction with conditional guarantees. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(4):1100–1126, 2025. doi:10.1093/jrssb/qkaf008.
- [54] T. Christensen, H. R. Moon, F. Schorfheide. Optimal Decision Rules When Payoffs are Partially Identified. *The Review of Economic Studies*, rdag017, 2026. Advance article, corrected proof. doi:10.1093/restud/rdag017.
- [55] P. R. Rosenbaum. *Observational Studies*. Springer, New York, 2002.